

Q. 10. Linear ODEs

(a) $\dot{x} = 7x$

Guess : $Ce^{\lambda t}$

$$\Rightarrow C\lambda e^{\lambda t} = 7Ce^{\lambda t}$$

$$\Rightarrow \lambda = 7, C \in \mathbb{R}$$

$$\therefore x = Ce^{7t} \quad \forall C \in \mathbb{R}$$

(b) $\dot{x} = -3x$

Guess : $Ce^{\lambda t}$

$$\Rightarrow C\lambda e^{\lambda t} = -3Ce^{\lambda t}$$

$$\Rightarrow \lambda = -3, C \in \mathbb{R}$$

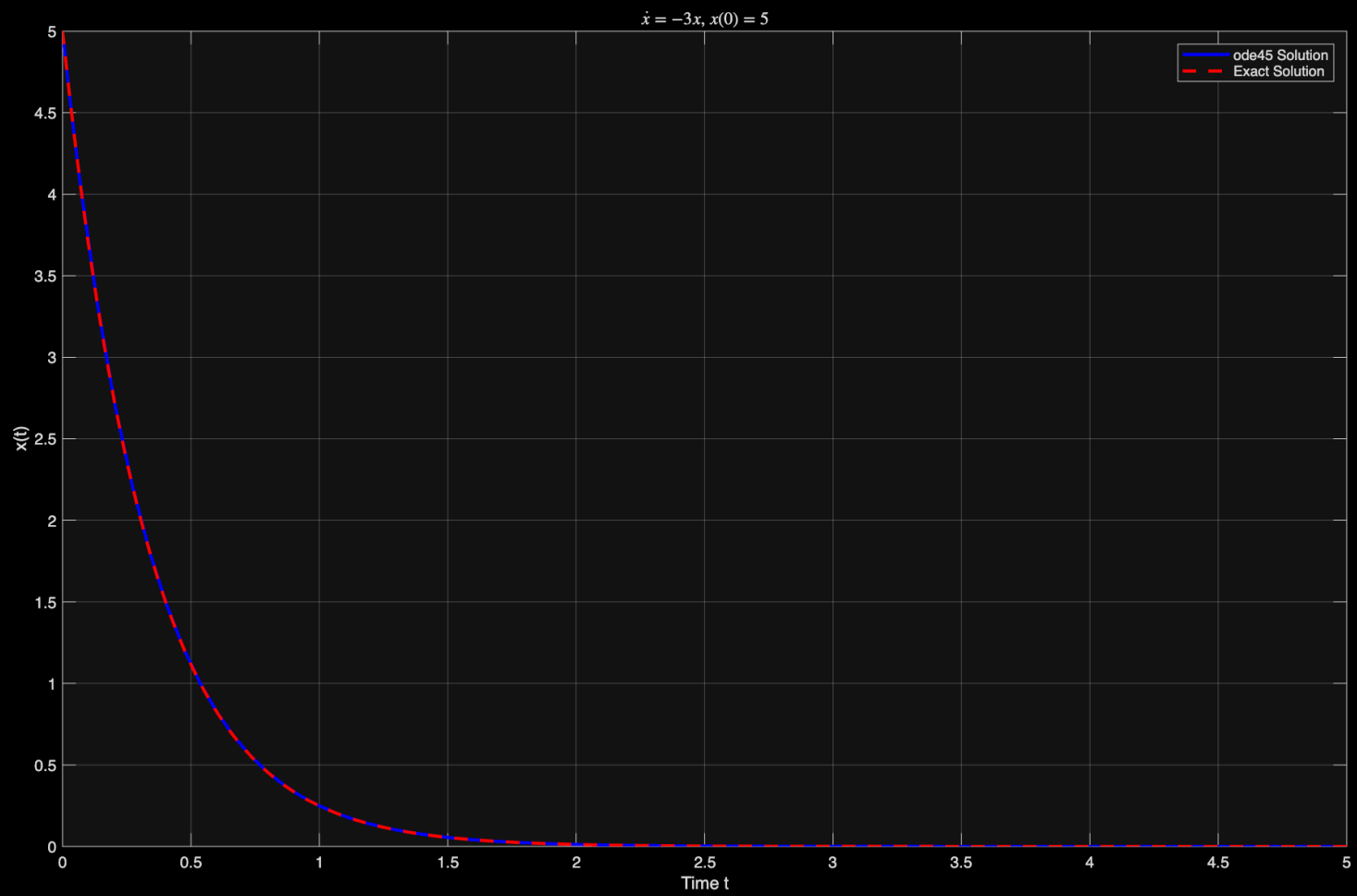
$$\therefore x = Ce^{-3t} \quad \forall C \in \mathbb{R}$$

IVP $x(0) = 5$

$$\Rightarrow 5 = Ce^{-3 \cdot 0}$$

$$\therefore C = 5$$

$$\therefore x = 5e^{-3t}$$



$$(c) \quad \ddot{x} = 9x$$

$$\text{Guess: } Ce^{\lambda t} = g$$

$$\dot{g} = C\lambda e^{\lambda t}$$

$$\ddot{g} = C\lambda^2 e^{\lambda t}$$

$$\Rightarrow C\lambda^2 e^{\lambda t} = 9Ce^{\lambda t}$$

$$\Rightarrow \lambda = \pm 3, \quad C \in \mathbb{R}$$

$$\therefore x = C_1 e^{3t} + C_2 e^{-3t}$$

$$\sinh(3t) = \frac{e^{3t} - e^{-3t}}{2}; \quad \cosh(3t) = \frac{e^{3t} + e^{-3t}}{2}$$

These are linear ODEs,

$\sinh$ and $\cosh$ are just linear combinations of x .

$$\therefore x = C_1 \sinh(3t) + C_2 \cosh(3t)$$

is a valid solⁿ.

or even

$$x = C_1 \sin(-3t) + C_2 \cosh(-3t)$$

$$(d) \quad \ddot{x} = -4x$$

$$\text{Guess: } g = A \sin(\omega t + \phi), \quad \dot{g} = A\omega \cos(\omega t + \phi)$$

$$\ddot{g} = -A\omega^2 \sin(\omega t + \phi)$$

$$\Rightarrow -A\omega^2 \sin(\omega t + \phi) = -4A \sin(\omega t + \phi)$$

$$\Rightarrow \omega = \pm 2, \quad A \in \mathbb{R}$$

$$\text{General sol}^n: \quad x = C_1 \sin(2t + \phi) + C_2 \cos(2t + \phi)$$

$$\text{for } C_1, C_2 \in \mathbb{R}, \quad \phi \in [0, 2\pi)$$

e) $\dot{z} = Az$ $z \in \mathbb{R}^{2 \times 1}$, $A \in \mathbb{R}^{2 \times 2}$

Guess: $z = e^{\lambda t} \cdot \vec{v}$ const. vec.

$$\Rightarrow \vec{v} \cancel{e^{\lambda t}} = A \cancel{e^{\lambda t}} \cdot \vec{v}$$

$$\Rightarrow \lambda \vec{v} = A \vec{v}$$

$\Rightarrow$ Solution is $e^{\lambda t} \cdot \vec{v}$ s.t. λ is the eigenvalue of A for any constant vector $\vec{v}$.

f) $\because A \in \mathbb{R}^{2 \times 2}$, no. of eigenvalues = 2
 $\quad \quad \quad \hookrightarrow \lambda_1, \lambda_2$

$$\therefore z = C_1 e^{\lambda_1 t} \cdot \vec{v}_1 + C_2 e^{\lambda_2 t} \cdot \vec{v}_2$$

g) IVP: $z(t=0) = z_0 \in \mathbb{R}^{2 \times 1}$

$$\Rightarrow \vec{z}_0 = C_1 e^{\lambda_1 \cdot 0} \cdot \vec{v} \Rightarrow C_1 = \vec{z}_0 / \vec{v}$$

$$\therefore \text{sol}^n: z = \frac{\vec{z}_0}{\vec{v}} e^{\lambda_1 t} \vec{v} = \underline{\underline{\vec{z}_0 e^{\lambda_1 t}}}$$

$$h) \quad A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \quad z_0 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\dot{\vec{z}} = A \vec{z}; \quad \text{Guess: } z = e^{\lambda t} \cdot \vec{v} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\Rightarrow \lambda e^{\lambda t} \cdot \vec{v} = A e^{\lambda t} \cdot \vec{v}$$

$$\begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 3v_1 + v_2 \\ v_1 + 3v_2 \end{bmatrix}$$

$$\begin{cases} 3v_1 + v_2 = \lambda v_1 \\ v_1 + 3v_2 = \lambda v_2 \end{cases} \quad \left\{ \begin{array}{l} \text{this is true for } \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ \text{where } \lambda_1 = 4 \end{array} \right.$$

$$\text{Also true for } \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \begin{cases} 3(1) + (-1) = 2 = 2(1) \\ 1 + 3(-1) = -2 = 2(-1) \end{cases}$$

where $\lambda_2 = 2$

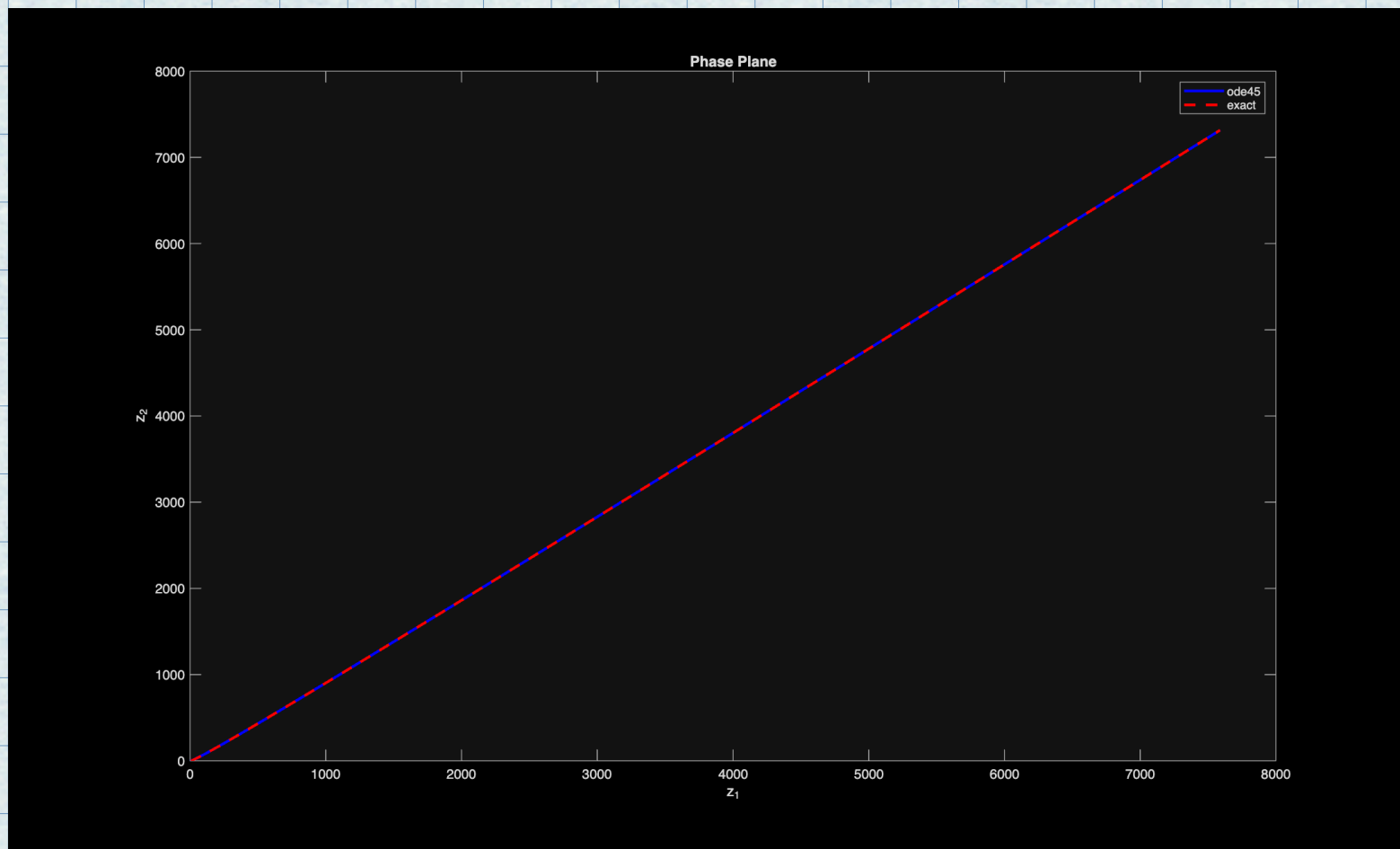
$$\therefore \text{sol}^n: \quad z = c_1 e^{4t} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{2t} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For IVP $z(0) = \begin{bmatrix} s \\ 0 \end{bmatrix}$,

$$\Rightarrow \begin{bmatrix} C_1 + C_2 \\ C_1 - C_2 \end{bmatrix} = \begin{bmatrix} s \\ 0 \end{bmatrix} \Rightarrow C_1 = C_2 = \frac{s}{2}$$

$$\therefore \text{IVP sol}^n : z = \frac{s}{2} e^{4t} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{s}{2} e^{2t} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(i)



(j)

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$z_0 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

